ADVANCED STATISTICS IN PRACTICE

Implications of Advanced Statistics in Business and Research

Turning data into evidence, decisions, and discovery

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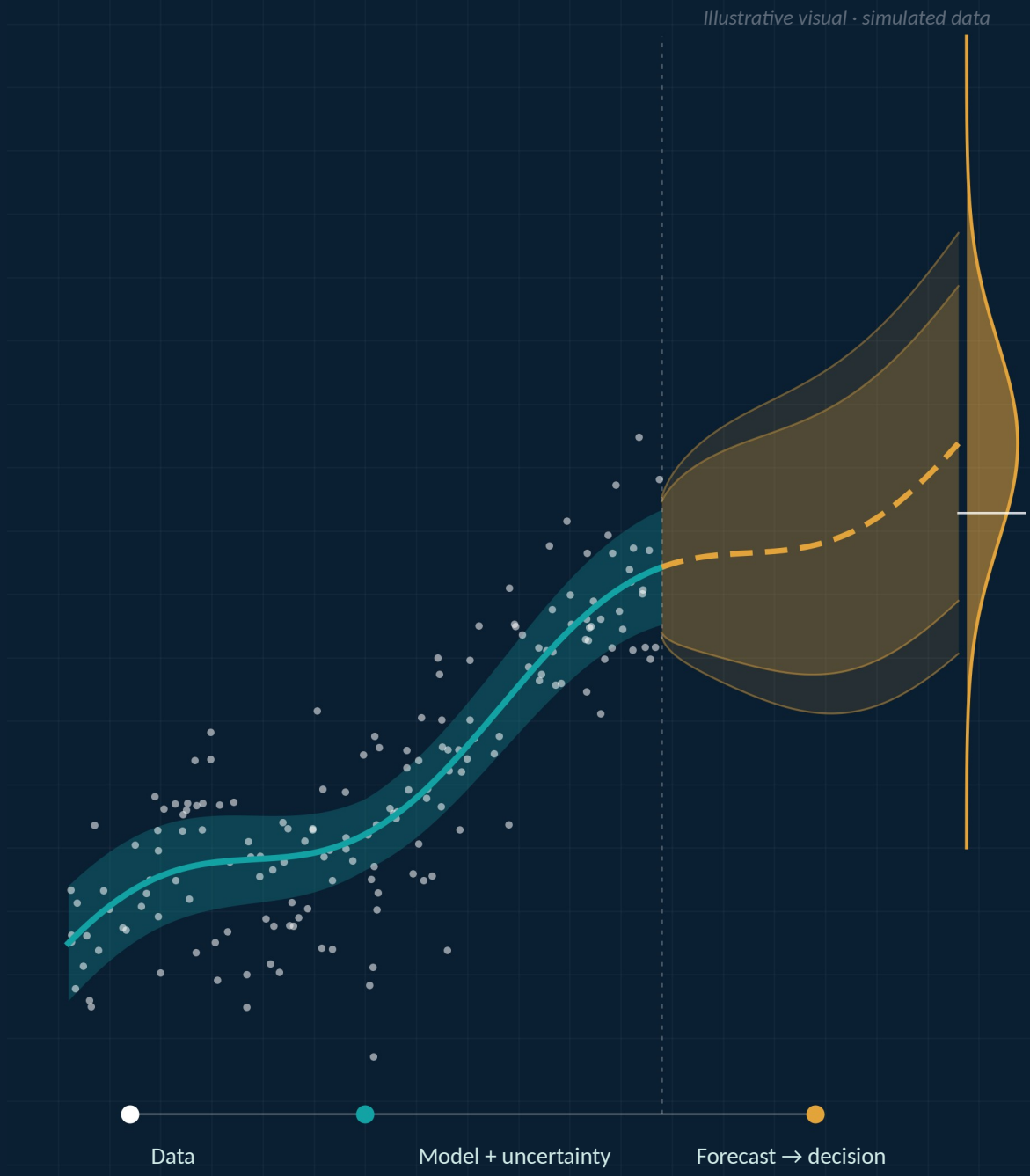
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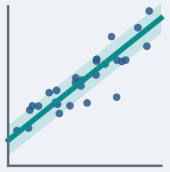


Statistics turns data into decisions

Advanced statistics helps us find real patterns, measure how sure we are, and choose with evidence instead of guesswork.

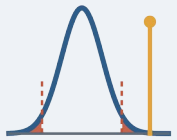
1 Collect data › 2 Find patterns › 3 Measure uncertainty › 4 Make decisions

Four key methods



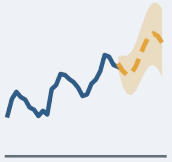
Regression

Shows how one factor affects another, e.g. price → sales.



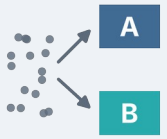
Hypothesis testing

Checks whether a difference is real or just chance.



Forecasting

Predicts future values, such as demand, with a range.



A/B testing

Compares two options fairly using random groups.

Better decisions in business and research

Business

- **Sales forecasting** → plan the right stock level
- **Customer segmentation** → target the right customers
- **Pricing** → find the price that earns most
- **Risk & fraud** → approve loans more safely

REAL CASE

Bing tested longer ad titles with an A/B test: revenue rose about 12% (≈ $100M a year in the U.S.).

Kohavi & Thomke (2017), Harvard Business Review

Research

- **Study design** → make comparisons fair
- **Sample size** → get reliable results
- **Confidence intervals** → show how sure we are
- **Replication** → build results others trust

REAL CASE

RECOVERY trial: dexamethasone cut 28-day deaths in ventilated COVID-19 patients from 41.4% to 29.3%.

RECOVERY Collaborative Group (2021), NEJM

ILLUSTRATIVE DATA hypothetical, not real company results

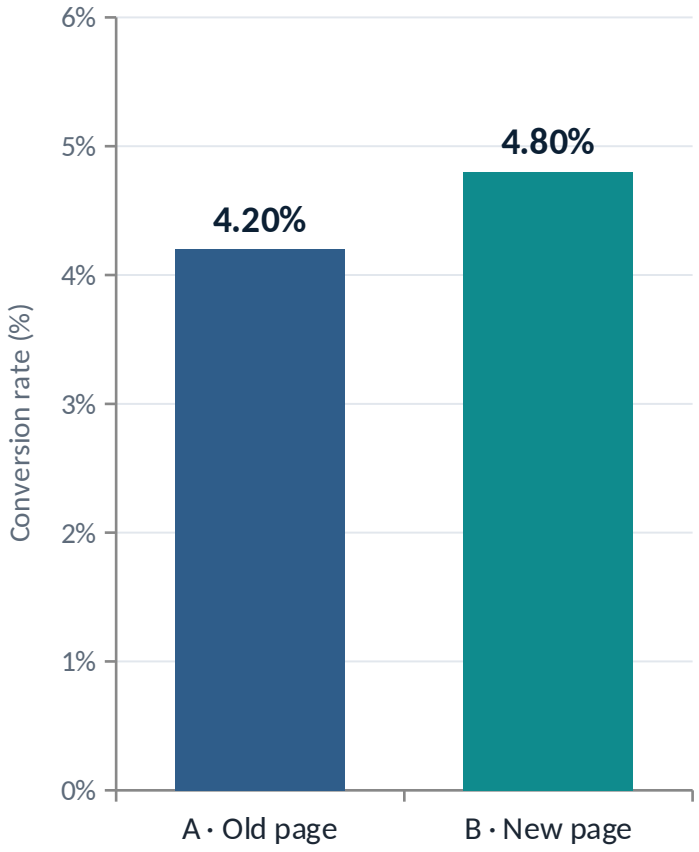
A/B test: does a new product page sell more?

1 Split
40,000 visitors at random: 20,000 see the old page (A), 20,000 the new page (B).

2 Measure
Conversion rate = orders ÷ visitors.

3 Test
Two-proportion z-test at the 5% level.

Visitors are assigned at random, so the only real difference between the groups is the page.



A: 840 / 20,000 · B: 960 / 20,000 orders

+0.60 pp
difference (B − A)

+0.19 to +1.01 pp
95% confidence interval

p = 0.004
very unlikely to be chance

Meaning The new page very likely works better, but the exact gain is uncertain.

DECISION
Launch page B and keep monitoring results.

Use statistics carefully and honestly

Statistics is powerful, but careless use can lead to confident and wrong decisions.

Bad or biased data

If the data is wrong or one-sided, the answer will be wrong too.

Correlation ≠ causation

Two things moving together does not prove one causes the other.

“Significant” ≠ important

A tiny effect can still be significant. Always check how big it is.

Privacy & fairness

Protect personal data and check that models are fair to every group.

WHAT ABOUT AI?

Machine learning is great at prediction, but a good prediction does not prove cause. Good practice: plan first, check the data, report results honestly.

Key takeaways

01

Data becomes evidence

Statistics turns raw data into results we can trust.

02

Better decisions

Testing ideas and measuring uncertainty beat guesswork.

03

Careful interpretation

How we read results matters as much as the method.

“The value of data lies not merely in collecting it, but in drawing reliable conclusions from it.”

Thank you · Questions?

References: Kohavi & Thomke (2017), Harvard Business Review 95(5) · RECOVERY Collaborative Group (2021), NEJM 384(8) · Wasserstein & Lazar (2016), The American Statistician 70(2) · Shmueli (2010), Statistical Science 25(3)